Notes on Navigation

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Contents

Chapter 1

Shapes, Lights and Sounds

For vessels at sea, the rules of the road are defined by the International Regulations for the Prevention of Collisions at Sea (IRPCS or COLREGS), a set of rules published by International Maritime Organisation (IMO) in 1972.

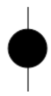
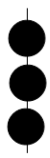
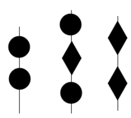
The COLREGS include a set of audio and visual signals that allow vessels to identify each other during the day (days shapes), at night (lights) and in restricted visibility/fog (sound signals).

1.1 Day Shapes

These are black geometric shapes, hung from a mast, to indicate the operational status of a vessel. The shapes used are designed to be recognisable from any viewing angle (ball, cylinder, cone, diamond).

Figure 1.1: Day Shapes

The following are some commonly used shapes:

Power underway	No shape required
Yacht sailing	No shape required
Tug Towing	
Fishing	
Trawling	
Limited by draught	
Restricted ability to manoeuvre	
Not under command	
Anchored	
Stopped	No shape required
Yacht motor sailing	
Aground	
Dredging	

Also of note is that a diving vessel will display the international code flag A as shown below:

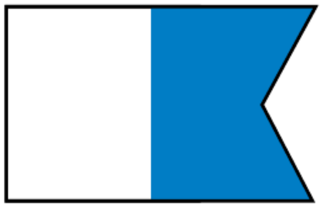


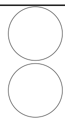
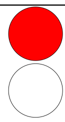
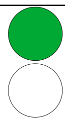
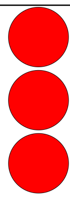
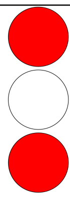
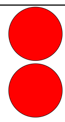
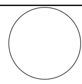
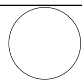
Figure 1.2: Flag A

1.2 Sound Signals for Vessels in Restricted Visibility

The following sound signals should be emitted by vessels in restricted visibility conditions every 2 minutes. The short blast should be about 1 second in duration while long blasts should be 4 – 6 seconds.

Power Underway	▬▬
Yacht sailing	▬▬ ▬ ▬
Tug Towing	▬▬ ▬ ▬
Fishing	▬▬ ▬ ▬
Trawling	▬▬ ▬ ▬
Limited by draught	▬▬ ▬ ▬
Limited by manoeuvre	▬▬ ▬ ▬
Not under command	▬▬ ▬ ▬
Anchored	▬ ▬▬ ▬
Stopped	▬▬ ▬▬
Yacht motor sailing	▬▬ ▬ ▬
Vessel being towed (if manned)	▬▬ ▬ ▬ ▬

Additional lights on the mast are used to indicate the operational status of the vessel. Some of these are listed below:

Power Underway	No mast lights, just navigation and stern white light
Yacht sailing	No mast lights, just navigation and stern white light
Yacht motor sailing	No mast lights, just navigation and all round white light
Tug Towing	
Fishing	
Trawling	
Limited by draught	
Limited by manoeuvre	
Not under command	
Anchored	
Stopped	

1.3 Navigation Lights

All vessels must display navigation lights between sunset and sunrise and also during times of restricted visibility.

All types of vessel should be equipped with side lights. The red light is always on the port/left side while the green is always on the starboard/right side.

A sailing vessel should display the sidelights along with a white light visible only from the stern. A power vessel should the sidelights along with a white masthead light visible from all directions. Vessels of more than 50m in length must display a second white masthead light. Note that a power vessel in this context could be a motor yacht.

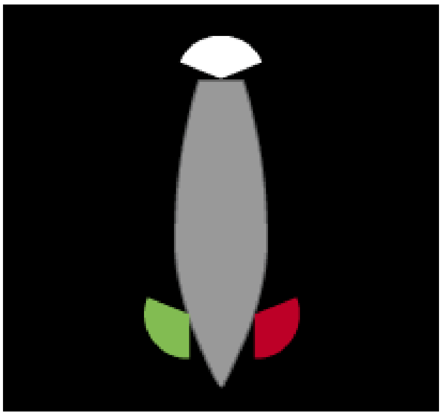


Figure 1.3: Sailing vessel, white light is only visible from the stern

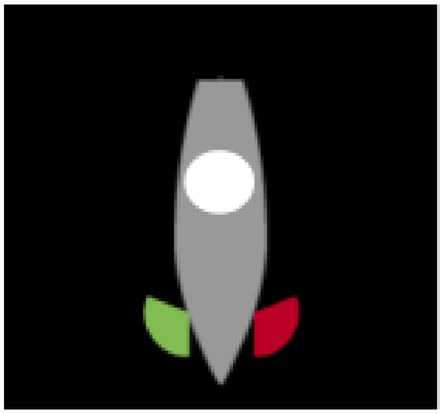
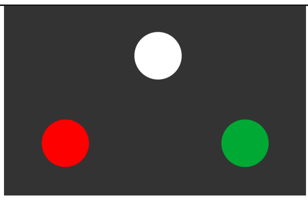
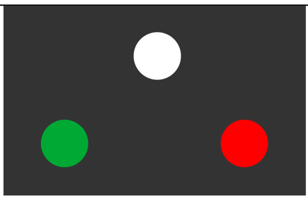
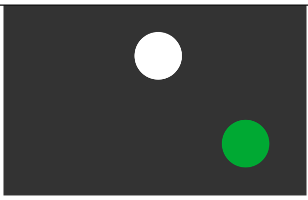
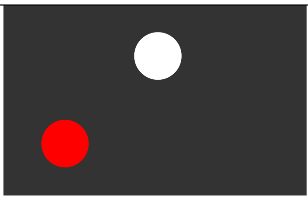
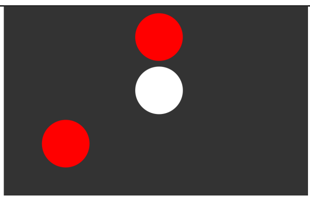
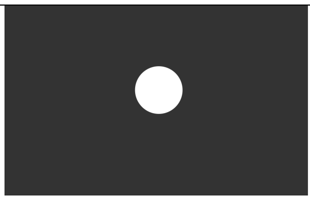
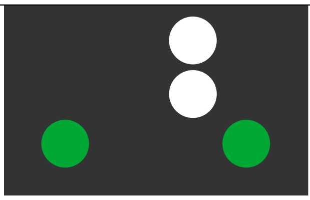


Figure 1.4: Power vessel, white light is visible from all aspects

Additional lights on the mast are used to indicate the operational status of the vessel. Some of these are listed below:

1.3.1 Example Silhouettes

	Sailing vessel underway, viewed bow-on (heading towards the viewer)
	Sailing vessel underway, viewed from starboard side
	Sailing vessel underway, viewed from port side
	Sailing vessel viewed from the stern or Motor vessel viewed from the stern
	Motor vessel viewed bow-on
	Motor vessel underway, viewed from starboard side
	Motor vessel underway, viewed from the port side
	Fishing vessel, viewed from the port side

	Anchored *or* Stopped *or* Sailing vessel viewed from stern
	Towing vessel and vessel under tow, viewed from starboard side

Chapter 2

Buoyage Marks

The International Association of Lighthouse Authorities (IALA) is an international organisation whose responsibilities include defining the Maritime Buoyage System (MBS) which includes:

- Cardinal Marks
- Lateral Marks
- Special Marks
- Safe Water Marks
- Isolated Danger Marks
- New Danger Marks

While the IALA system is used worldwide, the meaning of red vs. green lateral marks depends on whether you are in Region B (Americas, Philippines, Japan and Korea) or Region A (the rest of the world). Only region A conventions are used in this document.

2.1 Cardinal Marks

A cardinal mark is used to indicate the position the direction of safe water around a hazard. Cardinal marks indicate the direction of safety as a cardinal (compass) direction as opposed to lateral marks which indicate port and starboard relative to the direction of buoyage.

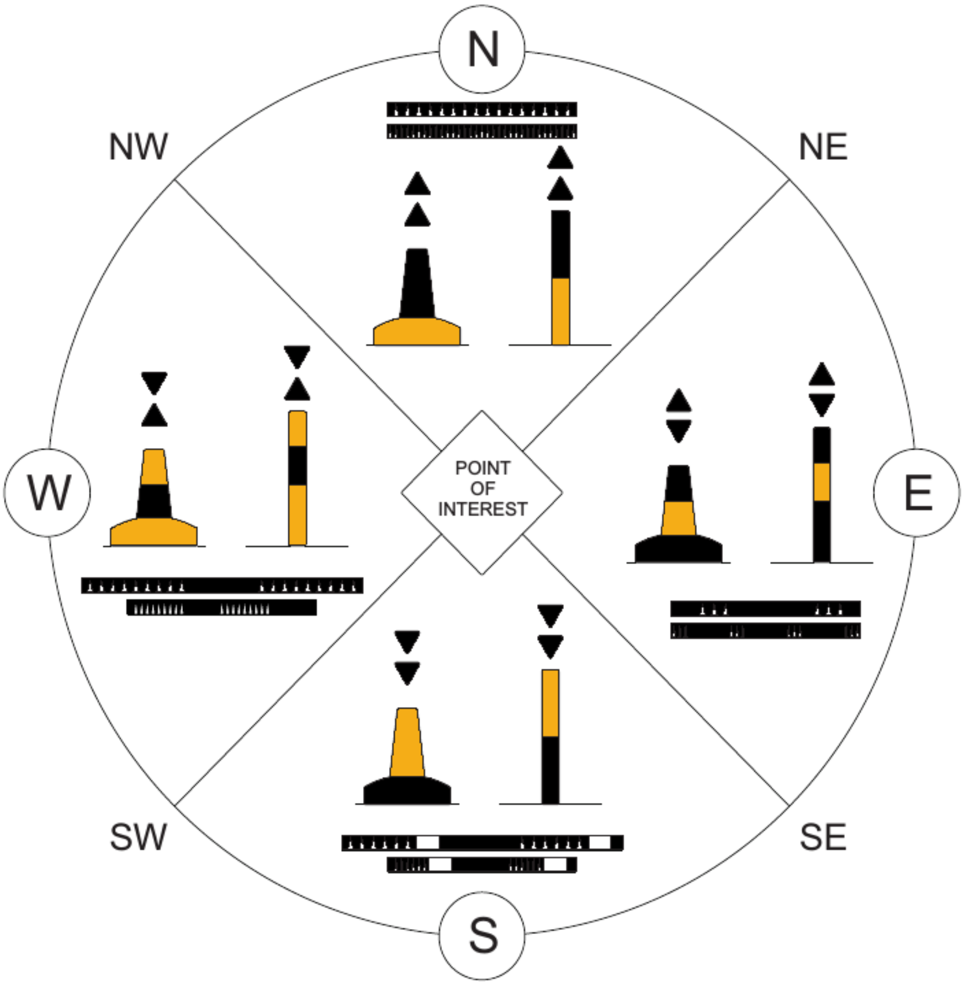


Figure 2.1: Cardinal Marks

The diagram above shows all of the characteristics of the cardinal marks. As indicated, they can be either pillars or spars. Each mark is distinguished by:

- The top-mark (the arrows on top)
- The colour
- The light pattern (if fitted)

2.2 Lateral Marks

A lateral mark is used to indicate the edge of the safe water channel. In Region A, for a vessel entering a harbour, the port side will indicated with red markers while the starboard side will be indicated with green markers.

While it might be obvious in many cases what is meant by "entering the harbour", to avoid confusion, charts showing lateral marks must also indicate the "direction of buoyage" as shown in the schematic below:

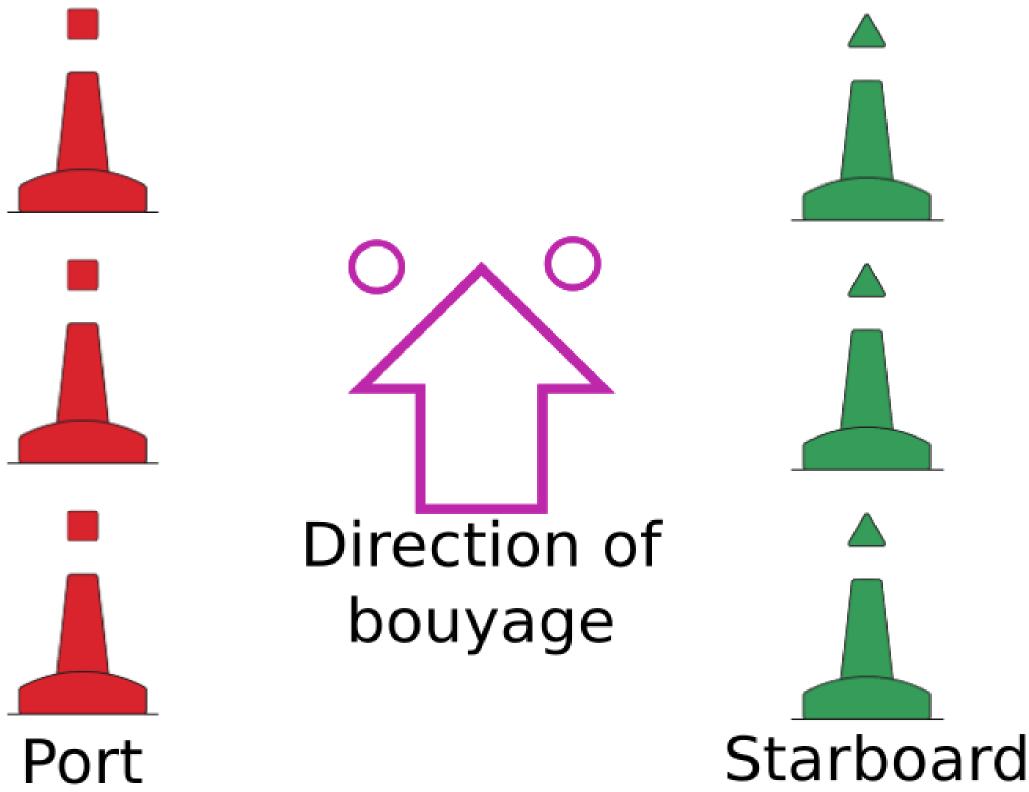


Figure 2.2: Lateral Marks

A more complete definition of the "direction of buoyage" is :

- The general direction taken by the mariner when approaching a harbour, river, estuary or other waterway from seaward or
- The direction determined by the proper authority in consultation, where appropriate, with neighbouring countries.

Lateral marks come in different shapes, not just the "spar" shapes shown above. Port hand marks can be cylinders, spars or pillars:

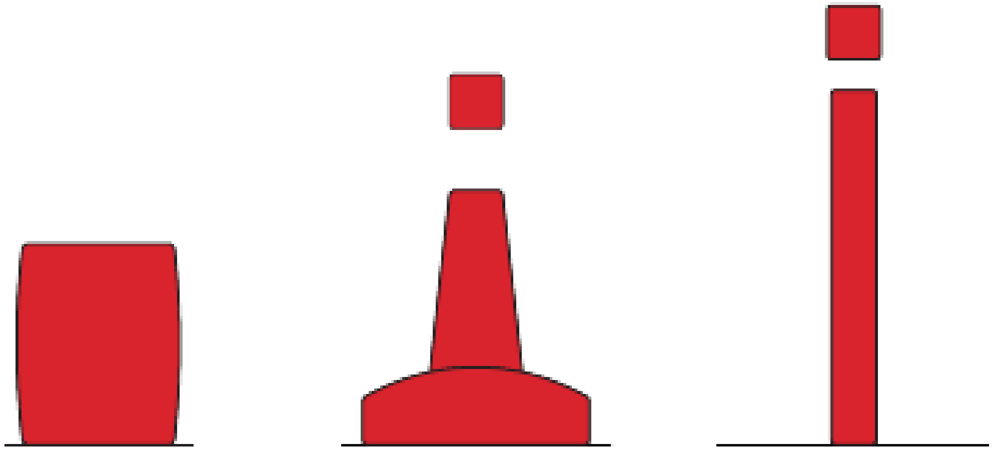


Figure 2.3: Port Marks

while starboard hand marks can be cones, spars or pillars:

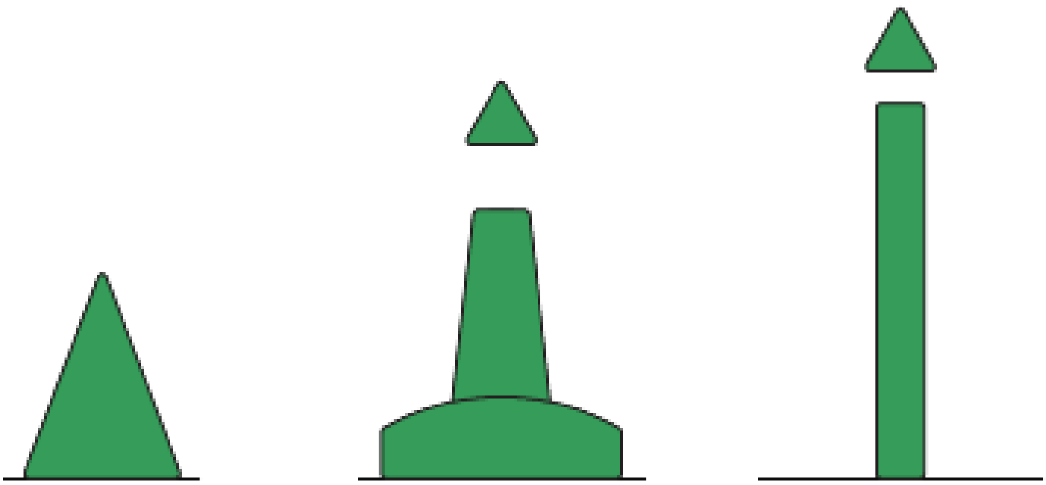


Figure 2.4: Starboard Marks

2.3 Preferred Channel Indicators

Where a channel splits, the preferred channel is indicated by modified lateral marks. Where the preferred channel is to port, the starboard mark will be modified to include a red stripe as below

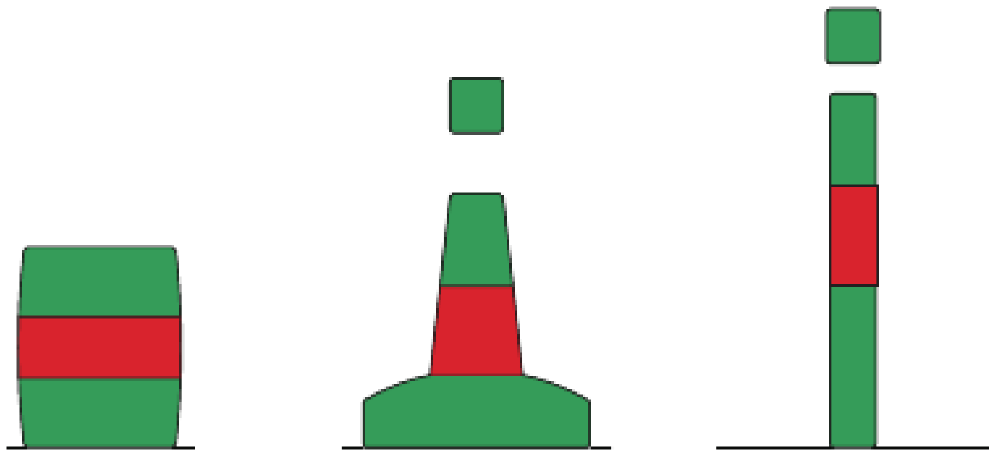


Figure 2.5: The starboard channel mark has been modified to indicate that the channel to the port is the preferred channel

Where the preferred channel is to starboard, the port channel mark will be modified to include a green stripe as below

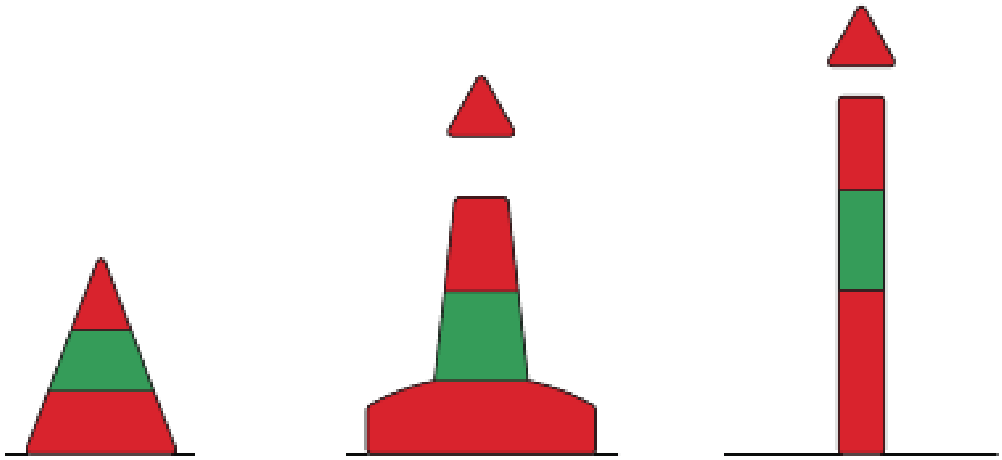


Figure 2.6: The port channel mark has been modified to indicate that the channel to the starboard is the preferred channel

In the example below, the preferred channel is on the port side. The starboard channel markers on the preferred channel have been modified to indicate this. The channel markers on the secondary channel are unchanged.

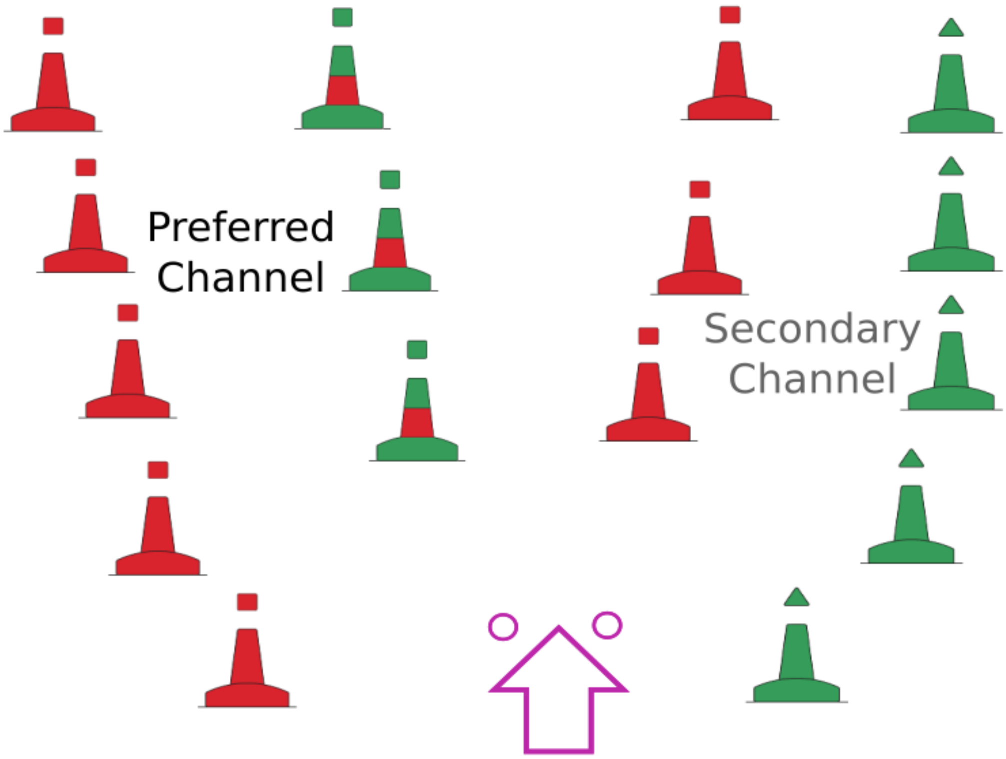


Figure 2.7: Preferred channel is port so starboard marker is modified in the preferred channel

2.4 Special Marks

Special marks are used to indicate specific features of interest. The nature of the special feature should be clear from the chart on which the mark appears. Special features might include:

- Ocean Data Acquisition System (ODAS) buoys
- Traffic separation schemes
- Spoil grounds
- Military exercise zones
- Cables or pipelines
- Recreation zones
- Boundaries of anchorage areas
- Structures such as offshore renewable energy installations
- Aquaculture

Special marks are yellow and may carry a top-mark in the form of an X. Special marks come in a wide variety of shapes as the only stipulation is that they must not conflict with shapes used for lateral water marks[1]

[1]If a special mark has the same shape as a lateral mark, there's a potential for it to be misinterpreted as such under conditions of poor visibility. For this reason, a special mark on the port side must not be conical as it could be mistaken for a starboard lateral mark. Similarly, a special mark on the starboard side must not be cylindrical. A special mark on the port side may be cylindrical since that would not conflict with the lateral marks.

2.5 Safe Water Marks

Safe Water marks serve to indicate that there is navigable water all round the mark. These include centre line marks and mid-channel marks. Such a mark may also be used to indicate channel entrance, port or estuary approach, or landfall.

Safe water marks may be either a sphere or else a spar or pillar with a spherical top-mark. They will always be coloured with red and white vertical stripes. The top-mark is a solid red colour.

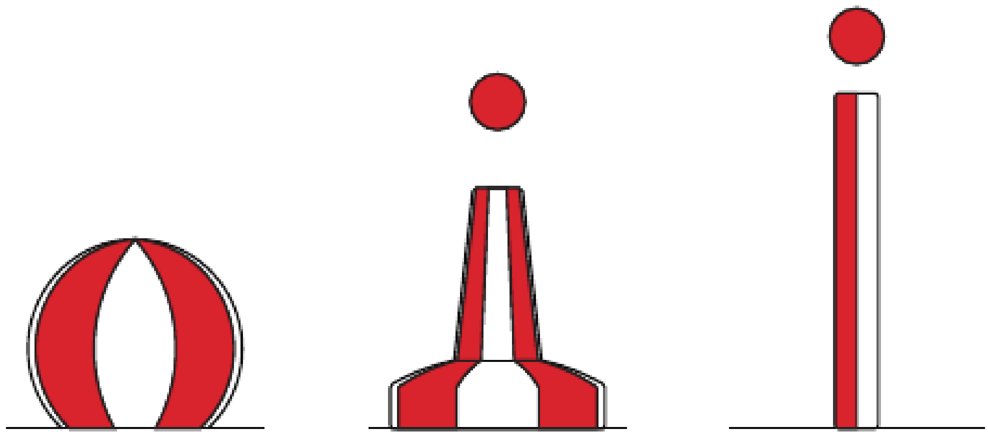


Figure 2.8: Safe Water Marks

2.6 Isolated Danger Marks

An isolated danger mark is placed on, or near to a danger that has navigable water all around it. The isolated danger mark has a distinctive top-mark consisting of two black spheres as shown below.

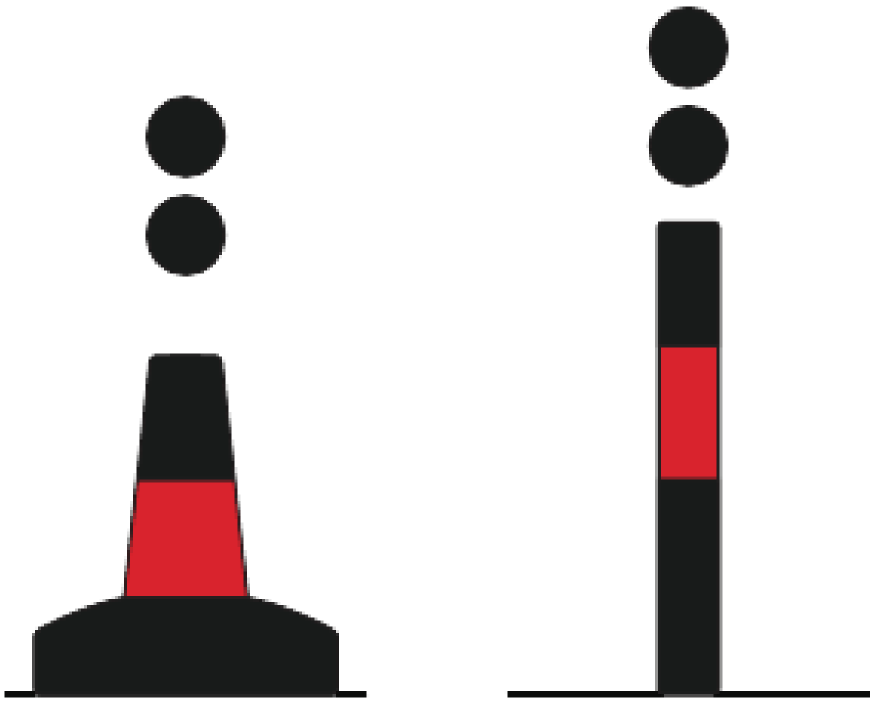


Figure 2.9: Isolated Danger Marks

2.7 New Danger Marks

The term “New Danger” is used to describe newly discovered hazards not yet shown in nautical documents. ‘New Dangers’ include naturally occurring obstructions such as sandbanks or rocks as well as man-made dangers such as wrecks.

The mark may either be a pillar or a spar and may have a yellow cross top-mark.

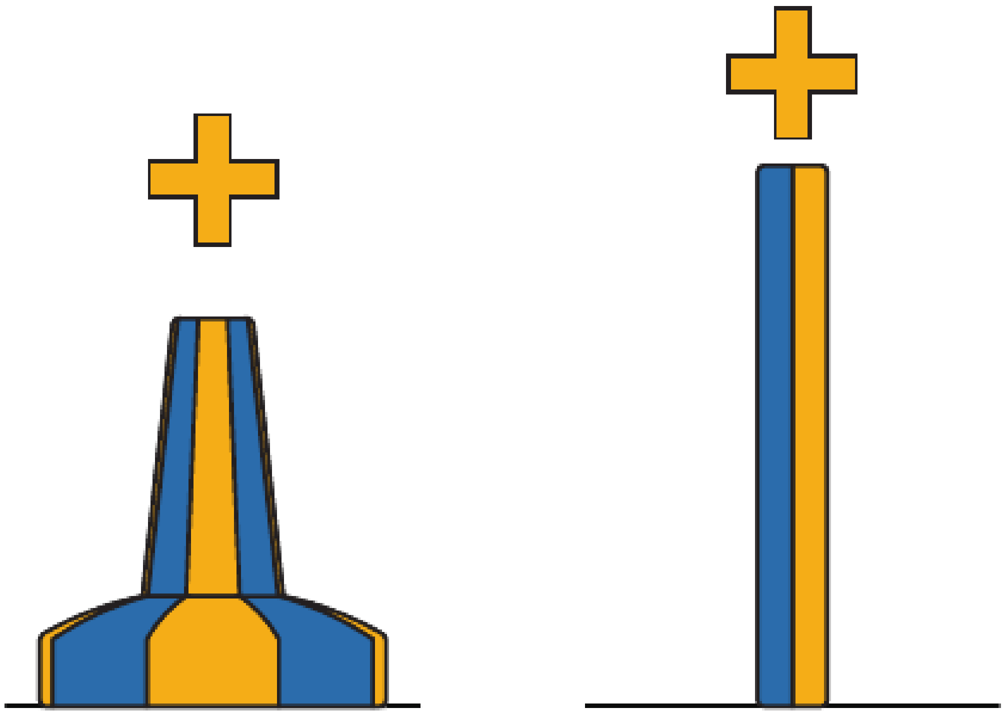


Figure 2.10: New Danger Marks

Chapter 3

Steering and Sailing Rules

3.1 Scope of the COLREGS

Part B of the COLREGS describes the steering and sailing rules. Regardless of the specific scenario or visibility conditions, the following two rules apply:

- Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.
- Every vessel shall at all times proceed at a safe speed so that she can take proper and effective action to avoid collision and be stopped within a distance appropriate to the prevailing circumstances and conditions.

The following are the main points covered in the steering and sailing rules

Sailing Vessels
When two sailing vessels are approaching, a vessel on a port tack should give way to a vessel on a starboard tack. A "port tack" means that the vessel has the wind coming from the port side.

If both vessels are on the same tack, the vessel to windward (that is, upwind), should give way to the vessel to leeward (downwind).

Motor Vessels Head On
When two power-driven vessels are meeting head on, each should move to starboard so that they pass port to port

Motor Vessels Crossing
When two power-driven vessels are crossing the vessel which has the other on their starboard side should keep out of the way

Overtaking
Any vessel that is overtaking should keep out of the way of the vessel being overtaken. This applies regardless of what type of vessels (motor or sail) are involved

Responsibilities
The full set of responsibilities are described in rule 18 of the COLREGS, part B. The main points are summarised below:

- A powered vessel should keep out of the way of a sailing vessel.
- A sailing vessel should keep out of the way of:
  - A vessel not under command
  - A vessel with restricted ability to manoeuvre
  - A vessel engaged in fishing

3.2 Traffic Separation Schemes

A traffic separation scheme (TSS) is used to regulate the direction of shipping in busy areas to avoid congestion. An example TSS from the English Channel is shown below

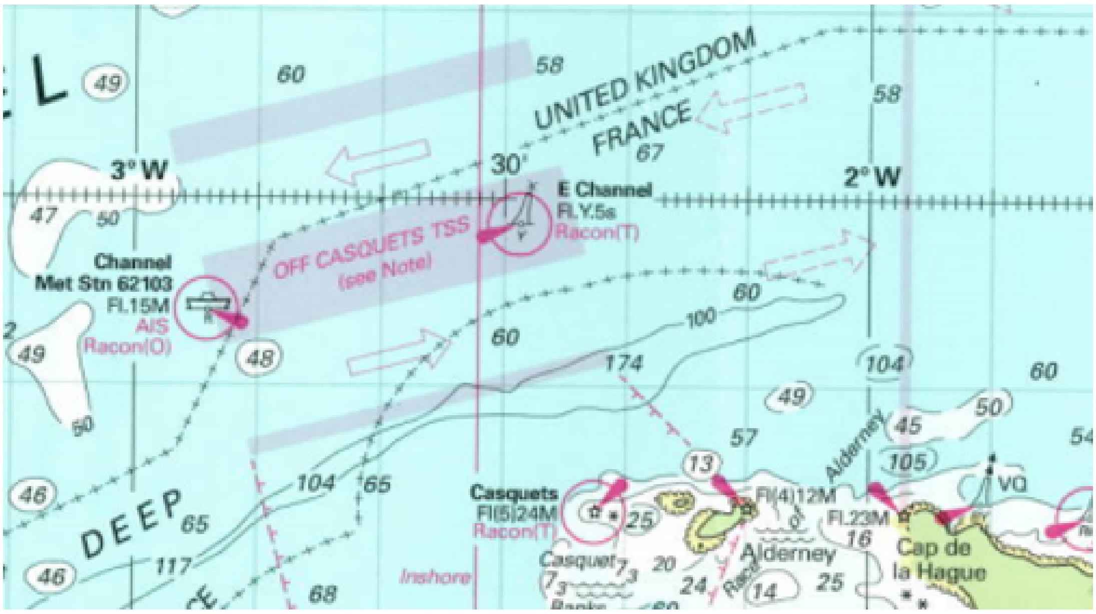


Figure 3.1: Traffic Separation Scheme

The direction of travel is indicated with the purple arrows and the area between these (in shaded purple) should be avoided.

It may occasionally be necessary for a vessel to cross a traffic separation zone. In this case, it's very important that the vessel should cross the zone heading at right angles in order to minimise the time spent crossing the traffic lanes.

This latter point can cause some confusion, it's important to remember that while the current could cause your vessel to deviate from the shortest course, you should not try to correct for this. You want to minimise the time that you spend in the traffic lanes, not the distance that you cover.

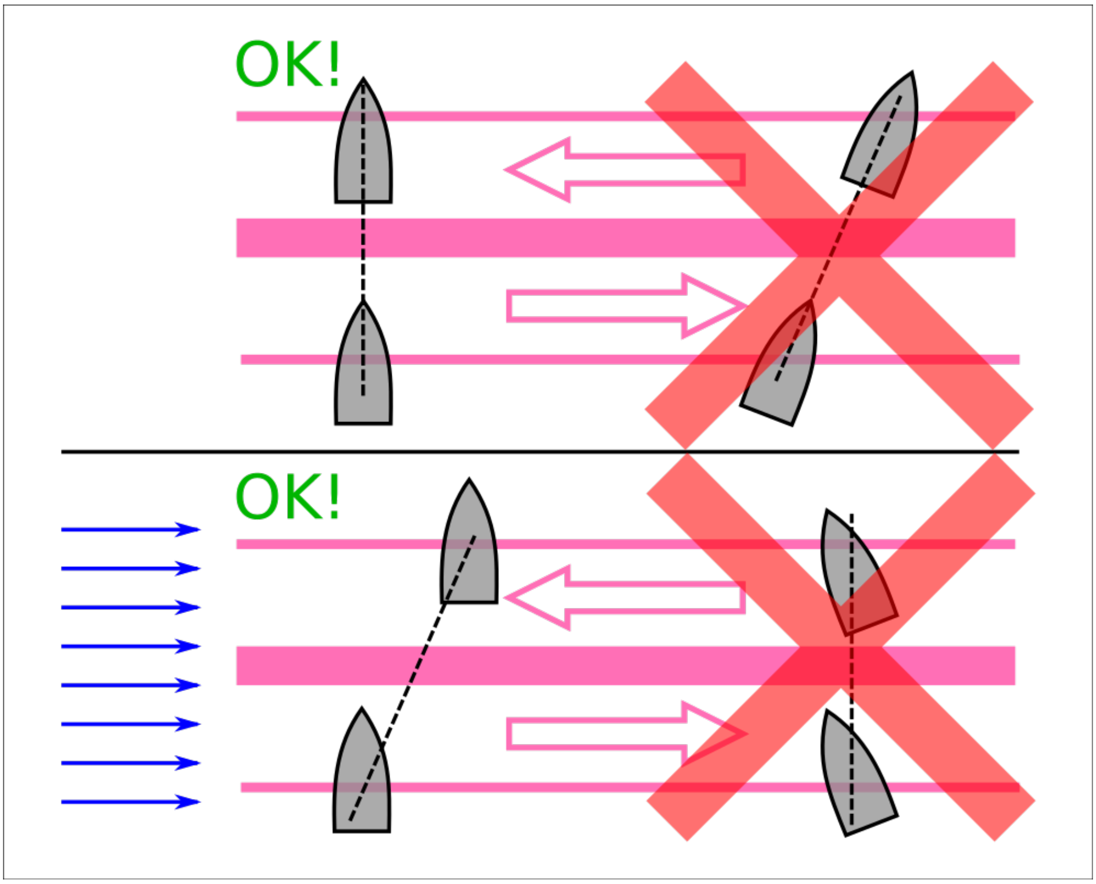


Figure 3.2: Crossing a Traffic Separation Scheme

Chapter 4

Charts and Tides

Below is an extract from an admiralty chart covering an area around Dunstaffnage in western Scotland.

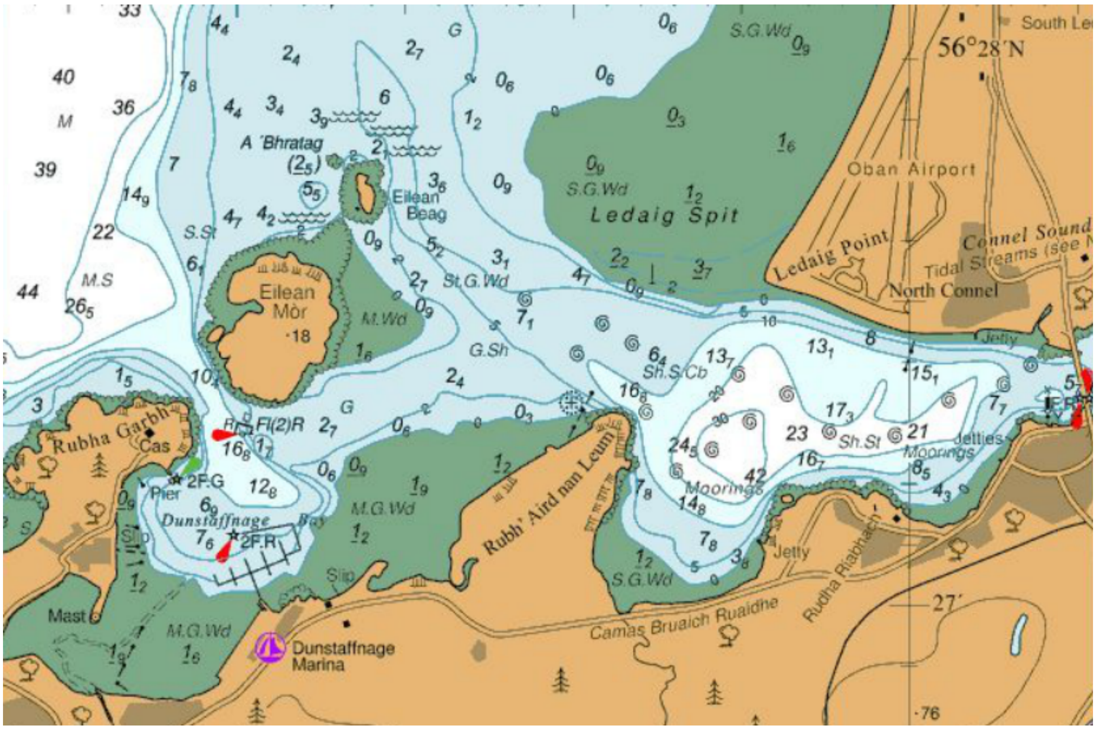


Figure 4.1: Example Chart

The areas in shades varying between blue and white represent the sea and the depth is shown at various points. In the example below, the depth at the indicated point is 2.4 metres (the subscript value represents decimetres, that is, 1 tenth of a meter)[1]

Figure 4.2: Depth Indicator

The actual depth at a particular time usually depends on the tide so a decision must be made as to

[1]In this particular chart, depths are in meters but some charts (especially those published in the US) use either feet or fathoms (a fathom is 6 feet).

which depth to indicate on the chart. A number of different levels might be chosen, for example, Lowest Astronomical Tide (LAT) or Mean Lower Low Water (MLLW). This level is called the chart datum and most charts (outside of the United States) use LAT as the chart datum.

Since LAT represents the lowest possible tide, the value shown indicates the minimum depth that can be expected at the marked location. Tide tables usually indicate the height above chart datum so the actual height you should expect to encounter is the value shown on the chart plus the height of the tide at the time you expect to be there. The areas shown in green represent areas that might be above or below the waterline depending on the tide. The numbers on the green area represent the height of the area above chart datum. This value is referred to as the drying height.

In the example below, if the tide were at its lowest possible level (LAT), the green area would be 1.2 meters above the level of the sea.

Figure 4.3: Drying Height

Note that the numbers indicating drying height have an underline below the meters value. This is to distinguish them from depth indicators.

- At a tide height of 0.5 meters, the green area would be 0.7 meters above the level of the sea.
- At a tide height of 1.6 meters, the green area would be below water, to a depth of 0.4 meters.

The areas shown in brown represent land, these areas are expected never to be under water.

4.1 Tidal Heights

Apart from Lowest Astronomical Tide (LAT), there are some other key heights that are important to know about. These are summarised in the diagram below:

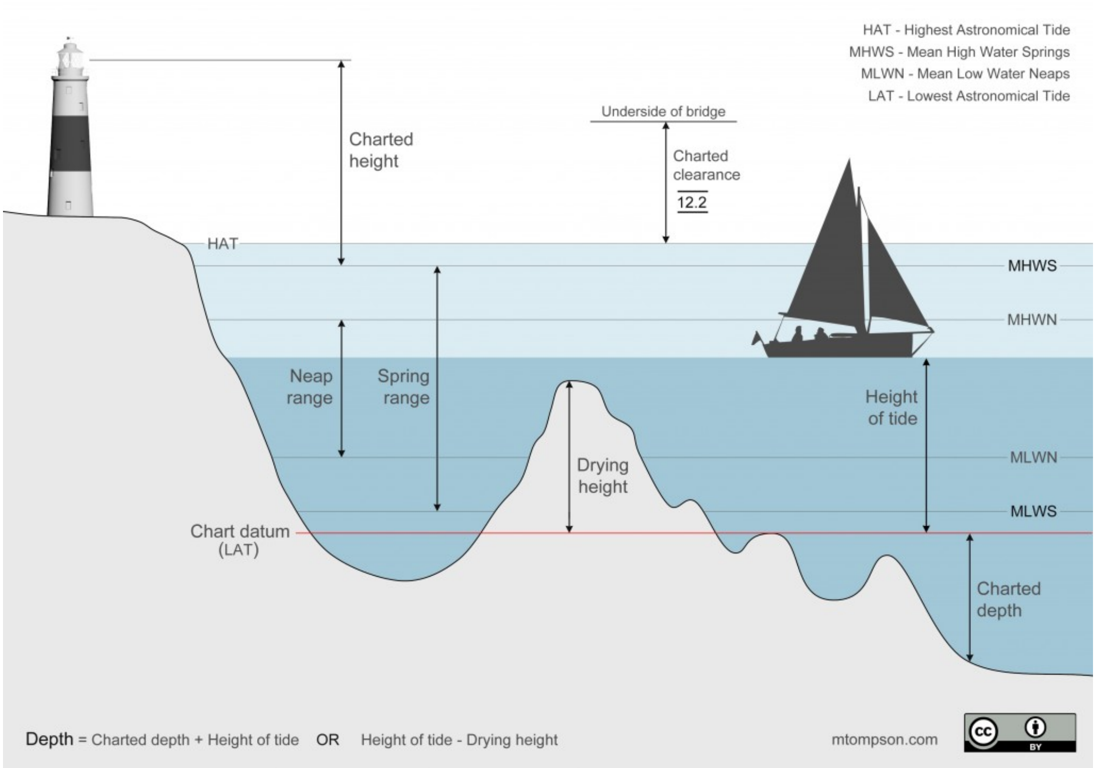


Figure 4.4: Tidal Heights

The depth at a blue or white area is the height of the tide + chart datum. For areas indicated in green, the depth is the height of the tide - the drying height.

4.2 Sample Tide Table

The table below shows the height of the tide at North Wall in Dublin during the month of April 2018. The times are adjusted for daylight savings time and the heights shown are in meters above LAT.

Tide at North Wall Dublin – April 2018											
	Time HH:MM	**Height metres**		**Time HH:MM**	**Height metres**		**Time HH:MM**	**Height metres**		**Time HH:MM**	**Height metres**
01 Sun	00:53	3.93	**02 Mon**	01:24	3.90	**03 Tue**	01:57	3.86	**04 Wed**	02:33	3.80
	06:29	0.45		07:06	0.44		07:42	0.49		08:23	0.59
	13:11	4.13		13:45	4.07		14:22	3.97		15:03	3.84
	18:54	0.25		19:30	0.35		20:09	0.50		20:48	0.70
05 Thu	03:13	3.71	**06 Fri**	03:57	3.59	**07 Sat**	04:45	3.43	**08 Sun** ◑	05:46	3.27
	09:07	0.75		09:57	0.93		10:53	1.14		11:55	1.32
	15:46	3.67		16:35	3.47		17:33	3.26		18:49	3.11
	21:31	0.92		22:19	1.16		23:15	1.40		-	-
09 Mon	00:20	1.58	**10 Tue**	01:33	1.64	**11 Wed**	02:49	1.52	**12 Thu**	03:45	1.31
	07:08	3.17		08:21	3.20		09:23	3.33		10:13	3.50
	13:04	1.41		14:33	1.36		15:24	1.19		16:08	0.98
	20:03	3.08		21:09	3.18		22:04	3.34		22:48	3.51
13 Fri	04:24	1.08	**14 Sat**	04:57	0.85	**15 Sun**	05:26	0.65	**16 Mon** ●	00:20	3.91
	10:54	3.67		11:28	3.82		12:00	3.95		05:57	0.48
	16:42	0.77		17:12	0.58		17:42	0.42		12:34	4.05
	23:23	3.66		23:52	3.79		-	-		18:15	0.32
17 Tue	00:52	4.00	**18 Wed**	01:30	4.04	**19 Thu**	02:12	4.04	**20 Fri**	03:00	3.97
	06:32	0.37		07:11	0.33		07:55	0.38		08:46	0.50
	13:12	4.11		13:54	4.11		14:42	4.05		15:33	3.94
	18:52	0.30		19:33	0.36		20:18	0.51		21:10	0.72
21 Sat	03:51	3.86	**22 Sun** ◐	04:50	3.72	**23 Mon**	06:06	3.59	**24 Tue**	00:33	1.33
	09:45	0.66		10:52	0.83		12:06	0.94		07:23	3.55
	16:29	3.78		17:35	3.61		18:54	3.49		13:24	0.95
	22:09	0.96		23:16	1.19		-	-		20:14	3.49
25 Wed	01:54	1.32	**26 Thu**	03:06	1.19	**27 Fri**	04:03	0.99	**28 Sat**	04:51	0.81
	08:42	3.63		09:50	3.77		10:48	3.91		11:37	4.00
	14:37	0.85		15:39	0.69		16:30	0.54		17:15	0.44
	21:27	3.58		22:28	3.71		23:18	3.80		-	-
29 Sun	00:01	3.85	**30 Mon** ○	00:36	3.85						
	05:33	0.68		06:11	0.60						
	12:19	4.02		12:55	3.99						
	17:56	0.42		18:33	0.46						
All times given are in local time											

4.3 Chart Symbols

Buoyage marks such as lateral marks, cardinal marks etc. are not usually printed in colour on navigation charts, instead the colour is indicated as an abbreviation as per the list below:

R	Red
G	Green
Y	Yellow
W	White
B	Black
Bu	blue

Combinations are indicated as a sequence of these letters, come common examples are shown below.

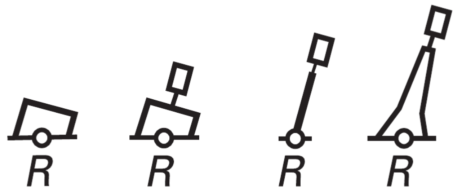
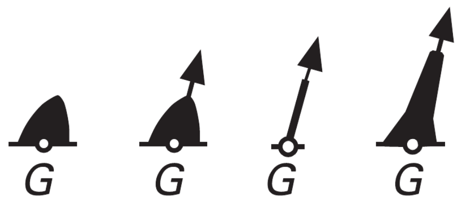
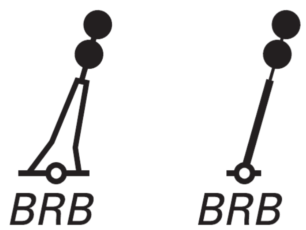
RGR	Red, Green, Red
GRG	Green, Red, Green
BRB	Black, Red, Black
RW	Red, White

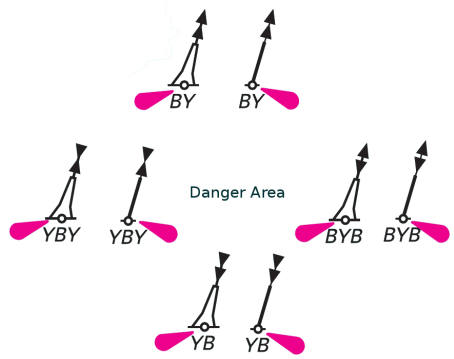
Similarly, where marks have lights, the pattern is indicated using a code. Some common examples are listed below.

F	Fixed	A continuous light
Occ	Occulting	A light with periodic darkness. The duration of the darkness will be shorter than the light
Iso	Isophase	Equal durations of light and dark
Fl	Flashing	Periodic lighting. The duration of the light will be shorter than the dark.
LFl	Flashing (Long)	A flash that lasts longer than 2 seconds
Q	Quick	Flashing but at higher frequency, 50 – 79 flashes per minute
VQ	Very Quick	Flashing at a frequency of 80 – 159 flashes per minute
UQ	Ultra Quick	Flashing at a frequency of 160 per minute or higher
M	Morse	Flashing in the morse pattern for the indicated letter.

The pattern should be followed by a colour, for example ***F R*** would indicate a fixed red light. With patterns other than fixed, a duration will also be required. For example, ***Fl G 5s*** would indicate that a green light flashes every 5 seconds.

The symbols used for various marks, along with their colour indicators are shown below:

Port hand lateral marks	
Starboard hand lateral marks	
Preferred channel to port	
Preferred channel to starboard	
Isolated danger mark	
Safe water marks	

Cardinal marks	

4.4 Other chart symbols

In addition to the depth markings, charts may also include indicators of the seabed composition. An earlier example used to indicate the drying height is shown again below. Below the depth indicator are the letters ***S.G.Wd***. This indicates that the seabed at this point is composed of Sand, Gravel and Weed.

Figure 4.5: Seabed composition S.G.Wd

A table of common seabed types is shown below

Symbol	**Description**	**Symbol**	**Description**	**Symbol**	**Description**
S	Sand	M	Mud	Cy	Clay
Si	Silt	St	Stones	G	Gravel
P	Pebbles	Cb	Cobbles	R	Rocks
Bo	Boulders	Sh	Shells	Sg	Seagrass
Wd	Weed (including kelp)	Co	Coral / Coralline algae		

One final symbol on the chart is the spiral symbol shown below. These indicate areas where the tidal stream produces rotating or disturbed water (eddies). These often form where strong tidal currents interact with an irregular seabed.

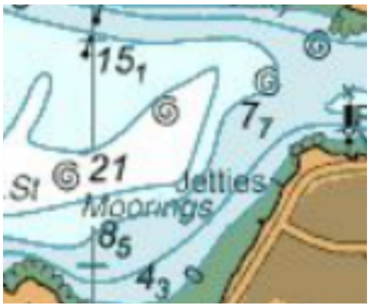


Figure 4.6: Eddies

Eddies are not necessarily dangerous but they should be noted when plotting a course as they may cause a significant difference between the apparent and actual course over ground.

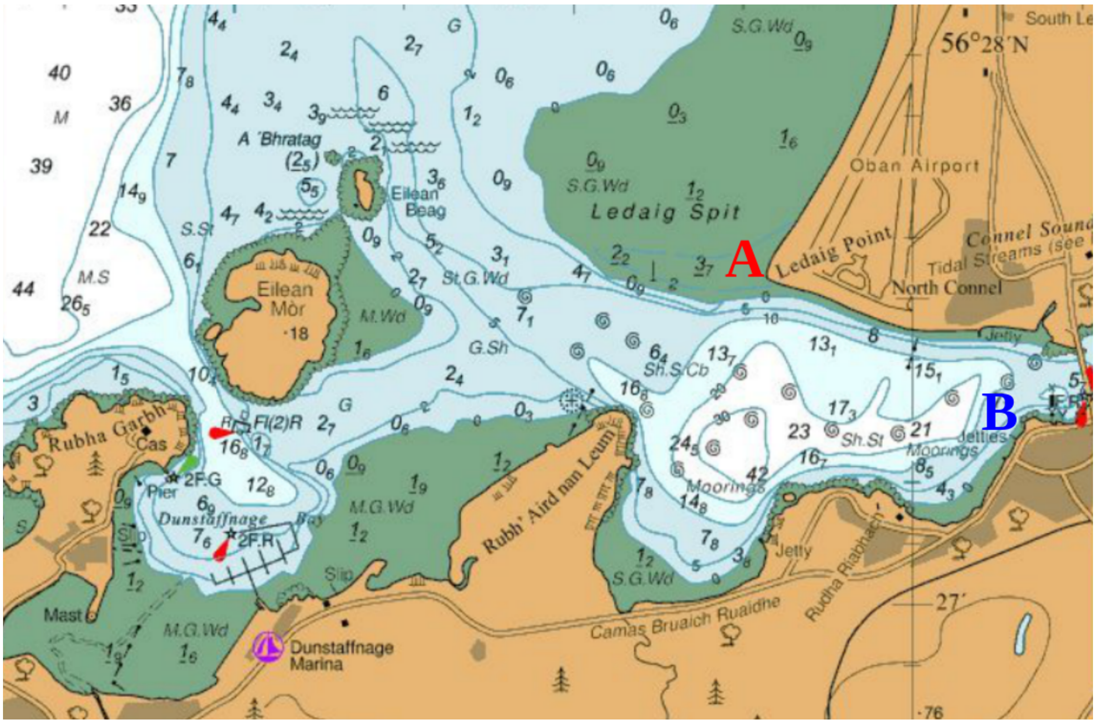
Chapter 5

Navigation

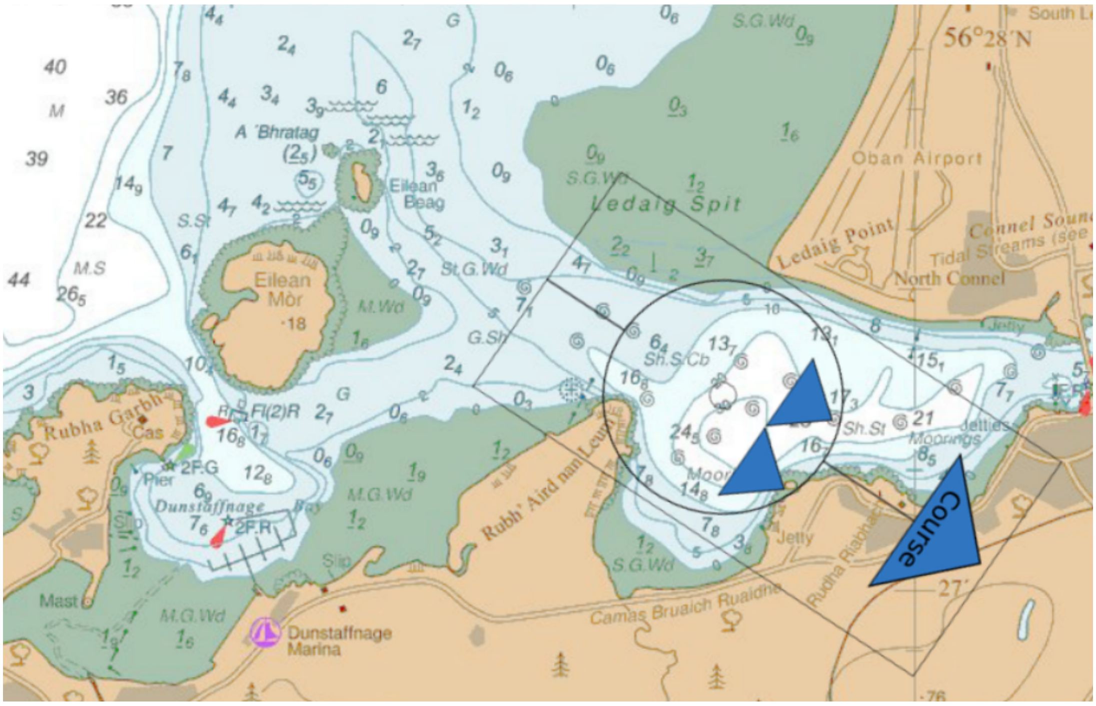
5.1 Determining Direction Using a Plotter

A plotter can be used to determine the direction of travel between two points on a chart. If you're planning a trip, you will need to apply some further adjustments to this figure before you can use it. The adjustments are explained in more detail in the sections that follow. For the moment, let's stick to direction as the crow flies.

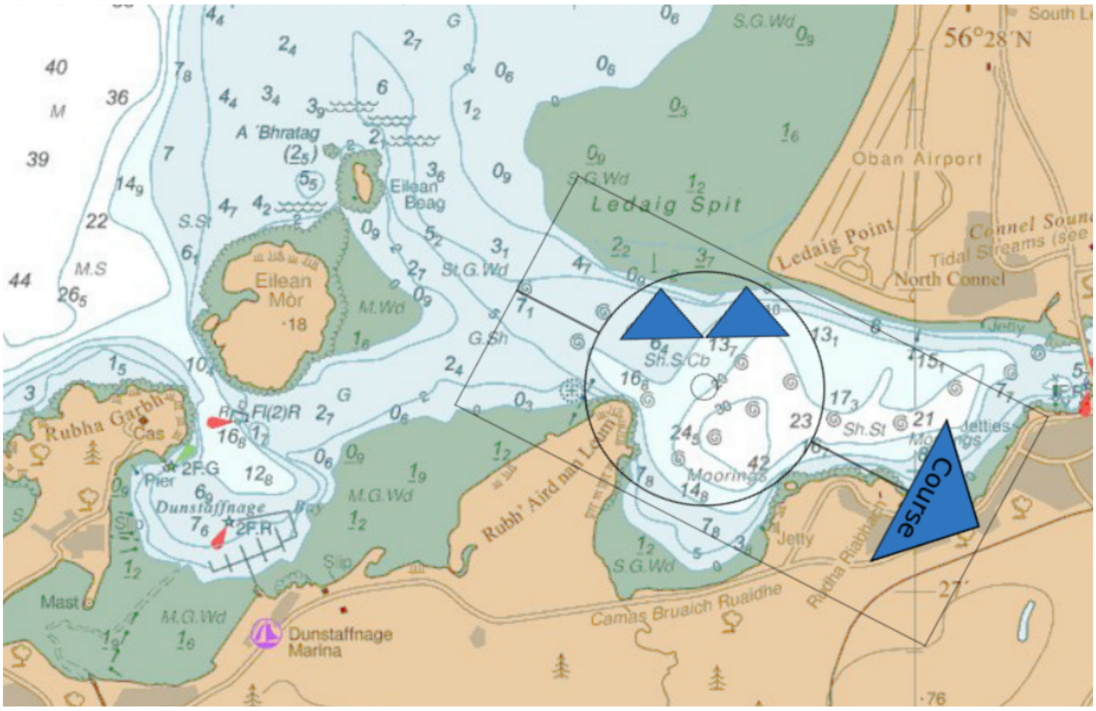
To take our first example, let's find the direction of travel when going from Ledaig Point (marked A) to the Jetties at point B



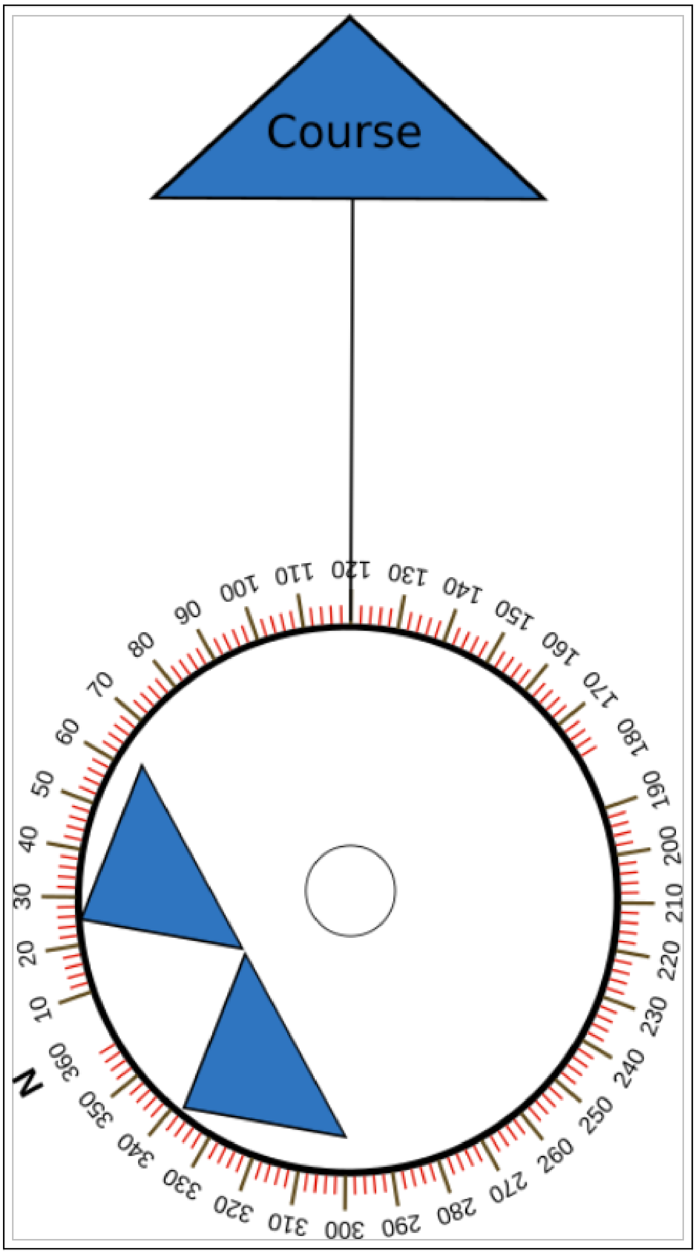
Step 1
Place your plotter on the chart with the edge of the plotter joining points A and B. Be careful of the direction, the blue arrow should point in the direction you want to travel (if the plotter is pointed in the opposite direction then all figures will be off by 180 degrees).



Step 2
Rotate the centre compass rose until the two blue arrows are pointing directly up



Step 3
Read the value off the plotter, it will be the number on the line that indicates your course direction. In this case, it looks like our heading will be 120 degrees.



5.2 Magnetic Variation and Deviation

It is very convenient for navigators that the earth has a magnetic field and that the North pole of this magnetic field is very close to the top of the axis of rotation of the earth (the geographic North). The end result is that you can hold out a compass and it will tend to point North.

While the above is mostly true, the geographic and magnetic Norths are not exactly aligned and if you don't take this into account, you will not be taking the course that you expect to take.

The difference between the magnetic and geographic North is called ***magnetic variation*** and the extent of variation depends on where you are on the Earth. Since the Earths magnetic field shifts over time, the extent of magnetic variation also changes over time although this is a very slow process, approximately 2 – 2.5 degrees per 100 years.

Furthermore, compass readings can be influenced by local factors. On a boat, this might be due to magnetised iron in the structure of the vessel. These localised effects are called magnetic deviation.

The degree of magnetic deviation can often depend on the direction the boat is facing. For this reason, the manufacturer may provide a compass deviation card for the boats steering compass. This card shows the extent of deviation for a given direction.

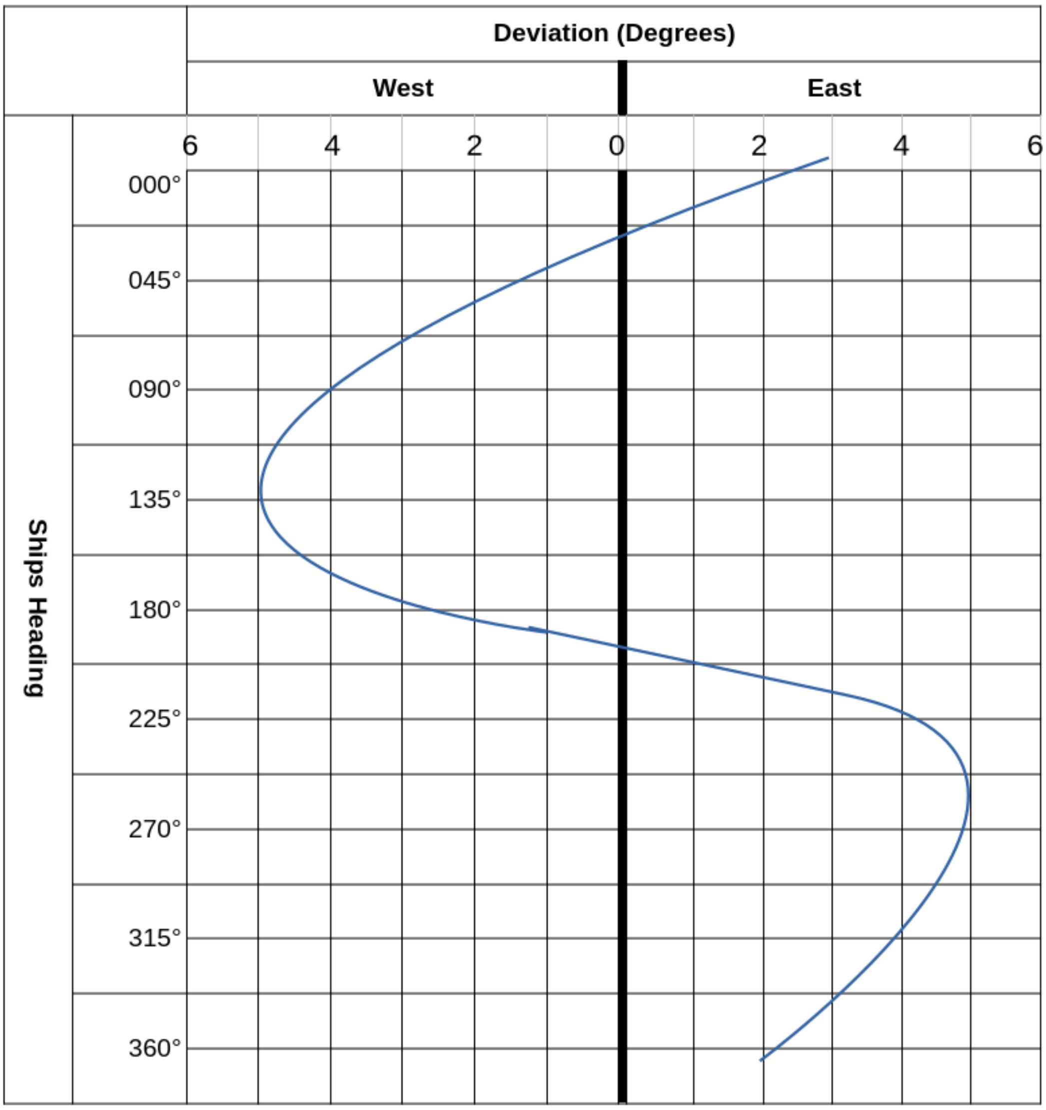


Figure 5.1: Sample compass deviation card for a boats steering compass. Depending on the boats orientation, the reading on the compass will be off, for example, by just under 2° east when the boat is pointing north (000°) and by 4° west when it is pointing east (090°).

In order to convert a reading from the chart into a reading on a compass, you must take both variation and deviation into account.

The mnemonic TVMDC can be used to help with this process. The letters stand for:

- True North
- Variation
- Magnetic
- Deviation
- Compass

In the previous example, our plotter indicated that our direction of travel should be 120°. This is the True North value but since our compass does not point to true North due to variation and deviation, we have to account for this.

Let's assume that the variation is 006° West. Based on the boat's deviation chart, our deviation will be approximately 4.9° West so we need to account for both.

True	**Variation**	**Magnetic**	**Deviation**	**Compass**
$120°_{T}$	006°W	$126°_{M}$	004.9°W	$130.9°_{C}$

Note that the directions have a subscript of T, M or C for True, Magnetic and Compass respectively.

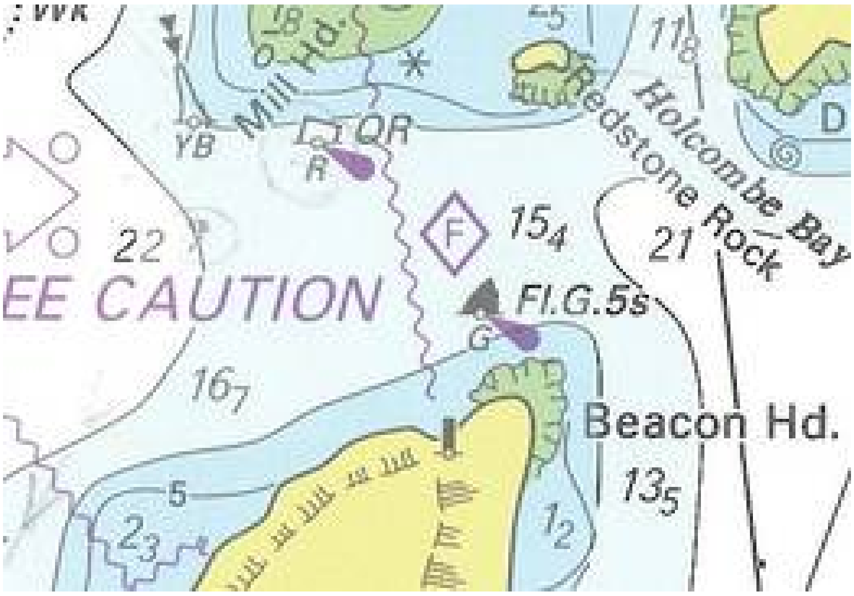
Why do we add the variation value?
Imagine that you want to travel directly North. On a clock face, this would be directly up to the number 12. In the example above, the magnetic pole is 6° West so it's not pointing at 12, it's actually pointing at 11:59 and if we follow the compass, that's where we'll end up. To get to 12, we should add 6° on to our compass reading.

The same logic obviously applies to the deviation value.

5.3 Tide Speed and Direction

A tidal table tells you at what time high water and low water will occur in a particular location. However, it does not tell you anything about the direction and speed of the tide, for that you need some additional information.

One way of conveying this additional information is with the use of ***Tidal Diamonds***. A tidal diamond is a purple diamond shape on a chart with a letter inside (for example, the letter **F** shown below).



Where tidal diamonds are used, the chart will include a lookup table so that the letter can be translated into speed and direction. An example is shown below:

Tidal Point F				
Hours		Dir	Sp	Np
Before High Water	6	248	0.8	0.4
	5	067	0.5	0.3
	4	068	1.9	1.0
	3	071	2.6	1.5
	2	069	2.3	1.3
	1	068	1.2	0.6
HW		**067**	**0.1**	**0.1**
After High Water	1	248	0.9	0.5
	2	247	1.4	0.8
	3	251	1.8	1.0
	4	253	1.7	1.0
	5	250	1.6	0.9
	6	249	1.2	0.7

Dir = Direction, Sp = Spring tide, Np = Neap tide

Imagine that you want to determine the tidal vector at point F at 14:30 on Saturday, 21st April. First, consult the tidal table for that day. Below is the required entry extracted from the table.

21 Sat	03:51	3.86
	09:45	0.66
	16:29	3.78
	22:09	0.96

According to this table, high water occurs at 16:29. We wish to know the tide at 14:30 so that's 2 hours before high water. The entry we need is highlighted in blue above, it tells us that at point F, 2 hours before high water, the direction on the tide will be 068° and the speed will be 2.3 kts at spring tide and 1.3 kts at neap tide.

Are we at a spring or a neap tide? According to the tide table, there's a spring tide on the 17^{th} and 18^{th} where the level reaches 4.11m. It then reaches a minimum of 3.49m on the 24^{th}. On the 21^{st}, we're about midway between a spring and a neap so we can pick the midway point between 2.3 kts and 1.3kts, that is, 1.8 kts.

In conclusion, our tidal vector will be 068° at 1.8 kts.

5.4 Estimating Position Taking The Tide Into Account

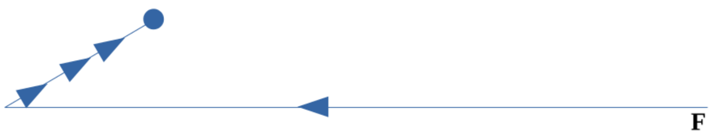
If we were to travel from point F, due West (270°) at a speed of 8 kts, we might expect that after one hour, our position would be 8 nm (nautical miles) to the West of point F. However, since the tide is pushing us in direction 068° (approximately Northeast) at a speed of 1.8 kts, we will end up slightly to the North and we will have covered less distance as well.

We have all of the information that we need in order to work out our estimated position after 1 hour, we can do this as follows:

Step 1
Draw a line on the chart starting at point F (the diamond below) and pointing in direction 270° (west). The length of this line should be 8 nm as measured on the chart. This line is called the water track and should be drawn with a single arrow as below.

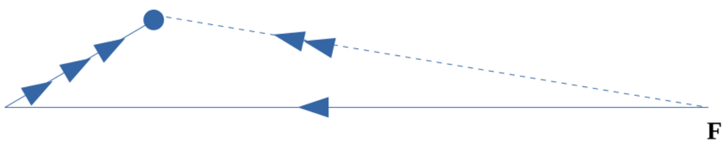
You can use a dividers to measure out 8 nm on the chart scale (be sure to use the vertical scale on either side of the chart).

Step 2
Draw the tidal vector starting at the end of your water track. The direction of the tidal vector will be 068° based on the workings above and the length will be 1.8 nm as measured on the chart vertical scale.



The estimated position after 1 hour is shown with the blue dot. Of course, in reality, the speed and direction of the tide won't be constant over the hour but this is still a reasonable estimate.

The line that joins the starting point to the blue dot is the ground track.

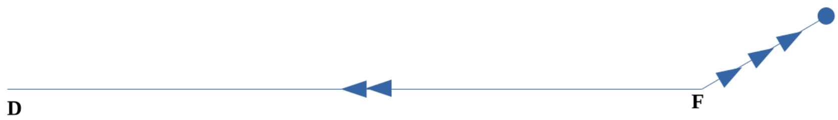


5.5 Plotting a Course To Steer (CTS)

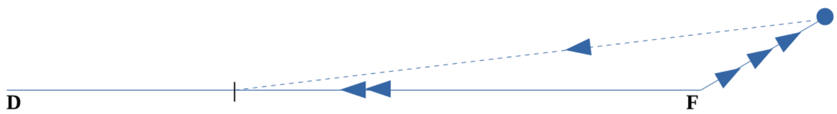
Continuing with the scenario above, let's imagine we want to reach a point 10 nm West of point ***F***. We know that if we simply point the vessel due West we will not reach our destination. To work out what direction we should take, we can take the following steps:

Step 1
Plot the course that we would like to take, this is our ground track. Measure 10 nm against the chart vertical scale using a dividers, this will be the length of our ground track. Here, our destination is marked with D.

Plot the tidal vector. This time, the tidal vector should begin at our starting position unlike our EP calculation where it begins at the end of the water track.



Step 3
Our boat has a speed of 8 kts so using a dividers, measure 8 nm against the chart scale. Then, place one end of the dividers on the blue dot and arc it so that it cuts the ground track. This point is marked below with a black vertical line and it's our estimated position after 1 hour. Don't worry if you overshoot your destination, if your destination is less than 1 hour away this will happen but the calculation works regardless.



> **Caution**
> You might be tempted to join the blue dot to the destination. This will not give you the answer that you need. You should fix the length and see where it cuts the ground track. This might be before or after your destination. Very rarely, it could be exactly at your destination but this will just be coincidence.

The dotted line joining the blue dot and this point is the water track. To determine the angle of the water track, place the edge of the plotter along this track (remember to point it in the direction of travel). Turn the compass rose until the arrows are pointing directly up and read off the value off the plotter. In this case, it should be approximately 266°.

This value is the true compass reading so it should properly be written as 266°$_T$. We now need to convert this into a compass reading by taking variation and deviation into account. We will use the same table as before. As with the original sample computation, we're assuming magnetic variation of 006° West.

First, we fill in the true reading and the variation. This gives us 272°_M.

True	**Variation**	**Magnetic**	**Deviation**	**Compass**
266°_T	006°W	272°_M	???	???

Next, take the value 272° and look up the compass deviation card. According to the card, our deviation at 270° is approximately 5° East. Since it's in the Easterly direction, we will need to subtract this value rather than add it.

True	**Variation**	**Magnetic**	**Deviation**	**Compass**
266°_T	006°W	272°_M	005° East	267°_C

So our course to steer (CTS) is 267°_C.

5.6 Calculating Speed Over Ground (SOG)

In the above example, we know that our speed over water will be 8 kts. What about our speed over ground? This is the effective speed that we will achieve when we take the tide into account.

To figure out our speed over ground, we need to work out the length of the ground track. We can do this by extending the dividers over this line. Then place the dividers against the vertical (latitude) scale on the chart, this will give you the distance covered in nautical miles. One minute on the vertical scale corresponds to 1 nautical mile (this is the original definition of the nautical mile). In this case, it should be approximately 6.3 nm.

Since we have based the calculations on travelling for 1 hour, this means that our effective speed (speed over ground) was 6.3 kts.

> Be careful to use the vertical scale (latitude) on the chart for measuring distances, this gives you the simple rule of 1 minute = 1nm. If you were to use the lines of longitude to calculate distance on a chart, you would need to do further calculations to translate the angle into a distance. This is because the lines of longitude are not really parallel and plotting them this way on a chart distorts distances..

5.7 Fixing Your Position

It is possible to determine your position by taking the bearing of landmarks that are identifiable on a chart. You can use these bearing to plot a track between your current location and the landmarks.

With a single bearing, you won't know your position, only that you are somewhere on the track that you plotted. With a second bearing, you can fix your position as the intersection between the two plotted tracks. However, to give reasonable certainty, you should always take at least three bearings.

When you take three bearings, you will likely find that the three lines don't intersect exactly at a single point, instead you will end up with a triangle. This smaller the triangle, the more precise your location.

You will typically take bearings with a hand compass at the front of the boat and as such, your readings should not be subject to the boats magnetic deviation. You will still need to take variation into account. In the example below, we will continue to assume magnetic variation of 006° West.

Firstly, holding your hand compass at eye level, you can see the flashing green marker; the compass reading is 262°. Adjusted for magnetic variation, this gives a value of 256°$_T$.

Turn the dial on your plotter to 256 and then place the plotter on the chart so that the edge touches the point on the green marker. Position the plotter so that the two arrows are pointing directly up. There is usually a grid on the plotter to help you get this exactly in line.

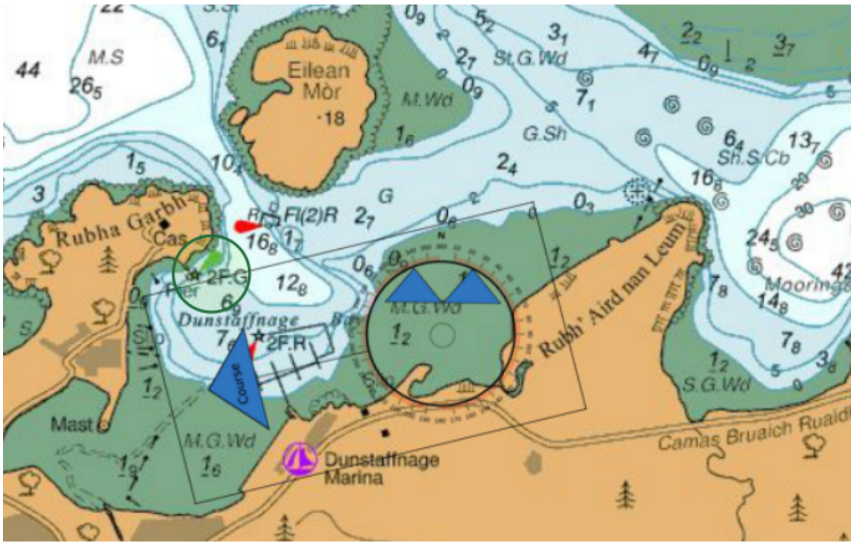
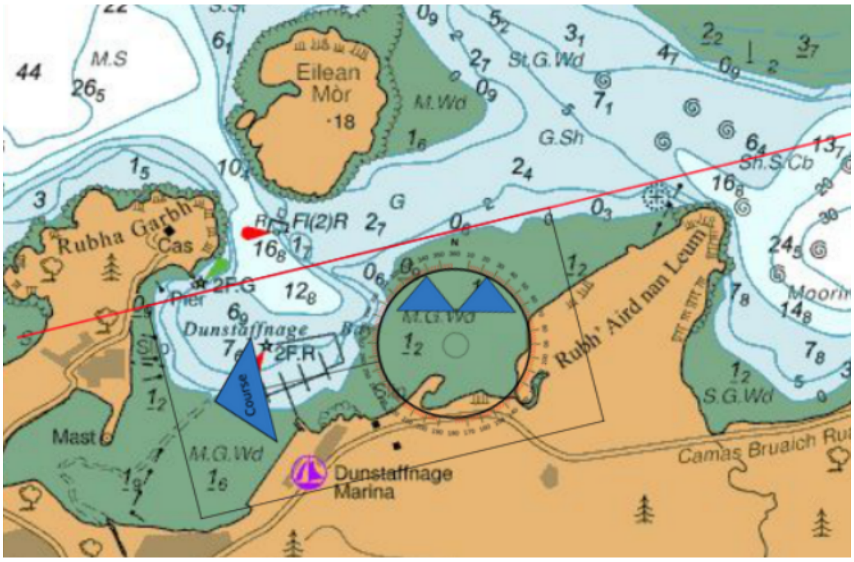


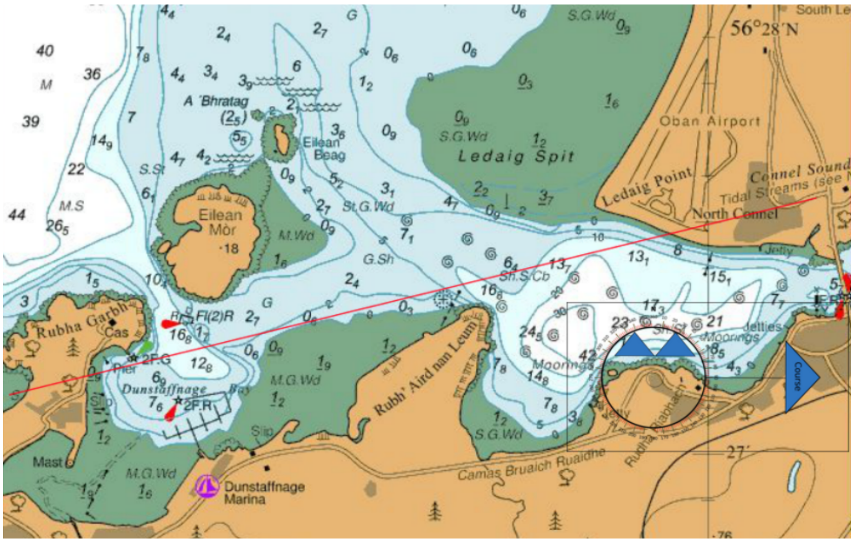
Figure 5.2: Plotter lined up with a flashing green marker, this is indicated with 2F.G

Now draw a line on your chart along the edge of the plotter that touches the landmark as shown in red below

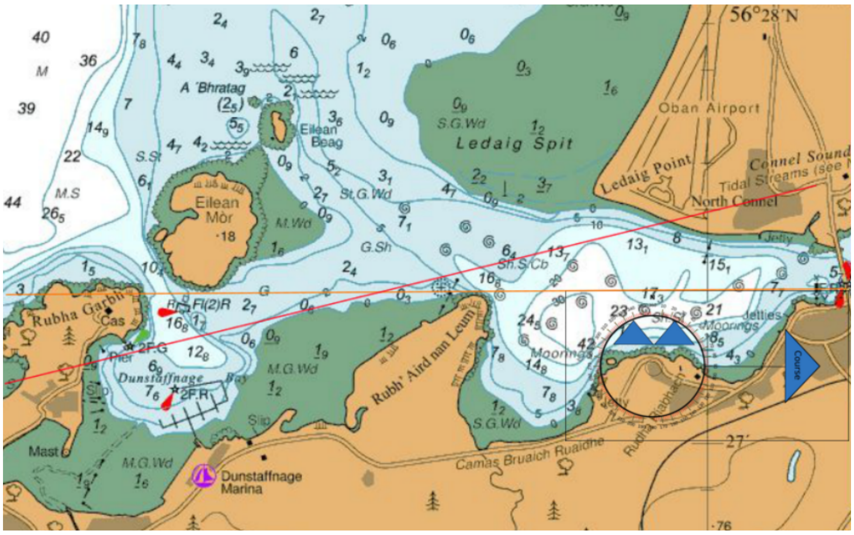


At this point, we know that we are somewhere along the red track. A second reading will fix our position and a third one will give us some extra certainty.

Our next bearing is the flashing red markers at 096°C, that is, 090°$_{\mathrm{T}}$. Rotate the rose on the plotter until it reads 090° and place it on the chart with the arrows pointing towards the flashing red markers.



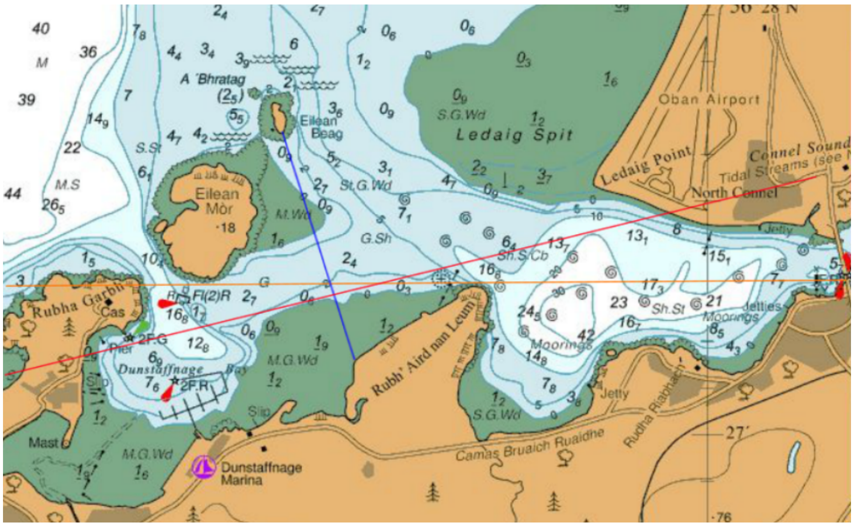
As before, draw a track along the edge of the plotter, this track is shown in orange below.



At this point, we can fix our position at the intersection between the red and orange tracks. However, a third bearing will give us a little more confidence.

If the bearing for Eilean Beag is 352°$_{C}$, corresponding to 346°$_{T}$, the as before, dial the compass rose to 346 and place it on the chart with the edge passing through your bearing point. Draw a track along this edge.

In an ideal world, all of you readings would be perfect and all three lines would intersect at a single point. In reality, it's likely that you will get a triangle as shown below.



The smaller this triangle is, the better. If you get a very large area, that suggests that you might have an error in your fix so it's worth checking your workings for possible errors.

Chapter 6

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